Four dimensional Yang–Mills research handoff guide

Research handoff

17 October 2026

What is being handed over

This bundle contains the current state of an attempt to complete a rigorous existence argument for a nontrivial four-dimensional Yang–Mills quantum theory. It includes a detailed research dossier, an annotated catalogue, all 39 proof notebooks as PDFs and editable LaTeX, and the underlying research and internal audit archive.

The central conclusion is precise: the working local ultraviolet chain reaches a positive actual determinant drift and arbitrarily long finite weak homogeneous histories. The final notebooks also establish exact full finite-measure identities and normalized local Wilson replacement bounds. The complete continuum existence theorem remains open.

The bundle is intended for a mathematician who can inspect the argument, challenge its dependencies, and continue at a specific unresolved theorem. It does not ask that reader to accept the collection as independently certified.

The three reading documents

1. **This start document** gives the result, the reading order, and the status boundaries.
2. **The research dossier** explains the pinned repository baseline, our additions, the exact scope of the strongest working results, primary literature interfaces, and the remaining existence program.
3. **The notebook catalogue** maps every notebook to its results, hypotheses, dependencies and limitations. It points to the accompanying correction register so superseded working notes cannot silently re-enter the proof.

At the bundle root, `index.html` gives clickable links to every PDF and source. `README.txt` provides a plain-text route. The machine-readable manifest records paths, sizes and SHA-256 hashes. No external application account is needed to read the PDFs or sources.

The substantial final stopping result

Notebook 39 finishes a restricted normalized replacement operation in the *actual bare Wilson measure*. Take any finite family of vertex-disjoint boxes of fixed lattice side. Condition on their common exterior, which includes every boundary edge. Their interior Wilson laws factor exactly.

On a good boundary, restrict a box law to its gauge-fixed conditional-minimizer window and divide by the exact probability of that window. On a bad boundary, keep the original law. The conditional tail bound ensures that every divisor is at least $1 - a_g$, with $a_g = Ce^{-cT^2} \leq 1/2$. The operation preserves the exterior marginal, gauge invariance and total raw partition mass. A positive expansion retains every removed configuration.

An explicit coupling gives, for a selected set A of interiors together with the whole exterior,

$$\|\mu_{(\mathrm{ext},A)} - \nu_{(\mathrm{ext},A)}\|_{\mathrm{TV}} = \mathbb{E}\left[1 - \prod_{i\in A}(1-p_i)\right] \le |A| a_g.$$

The convention is $\|\lambda - \lambda'\|_{\mathrm{TV}} = \sup_E |\lambda(E) - \lambda'(E)|$. Thus a local comparison does not deteriorate when distant boxes are added. The theorem also supplies discrepancy, cluster, entropy, bounded-source and connected-cumulant estimates.

Its restriction is essential: fixed boxes and their conditional minimizers are not yet the nonlinear coarse-average fibers and growing components of the multiscale construction. The theorem finishes a normalized local measure operation, not the existence problem.

What is already in the working chain

Local classical construction. Flat coercivity, constrained lifts, weighted inverse kernels, refinement control, nonlinear minimizers and localized responses near the trivial orbit.

Actual local quantum integration. The chosen composite ordinary map, exact constraint chart, independent retained window, Haar and pivot factors, gauge insertion, common analytic domain and supported expansion.

Finite local recursion. Total Ward extraction, exact finite totals and vacuum bookkeeping, controlled remainder induction and finite coupling shooting.

Positive actual coefficient. Under stated upstream inputs, Notebook 37 obtains

$$\beta(\mathcal{D}_{M,K}) = 22\alpha_g \log K + O(1), \qquad \alpha_g = C_H/(48\pi^2),$$

uniformly in preceding depth and the required joint structural limits.

Full finite-measure progress. Notebook 38 supplies exact positive history operators, normalized Wilson Peierls bounds, conditional box tails and gauge-invariant local defect bounds. Notebook 39 completes the normalized replacement just described.

The general $SU(N)$ statements in the new bare-measure lemmas do not extend every earlier $SU(2)$ construction input to all groups. Many local results depend on earlier modules; internal reviews are not external certification of that full chain.

The next theorem

The first major unresolved step is a *complete-density transfer theorem* for the actual chosen renormalization map. It must match a spatial good-component integral to the constrained background and retained coordinates already used, then bound normalized complementary components uniformly in their admissible exterior data, geometry and scale age.

The coupling extraction must belong to that same full-density construction. Weak numbers produced by a local recursion are not automatically the effective couplings of a different published large-field theorem. Bałaban's complete-density induction is a serious route once its actual hypotheses have been matched.

After this come uniform external observable-source estimates, a common continuum hierarchy, reflection positivity and full Euclidean covariance on the same limit, reconstruction, and a quantitative nontriviality and Yang–Mills identification argument. A physical mass gap is separate.

A practical reading order

1. Read the dossier's evidence rules and repository baseline, then its remaining theorem program. This prevents a local result from being mistaken for a continuum theorem.

2. For an independent finite-measure entry point, read Notebooks 38 and 39. Their bare Wilson estimates do not require the earlier determinant work.

3. For the local ultraviolet route, read Notebook 37 with its inputs from 36, 32 and 31. Use the catalogue to trace older dependencies rather than guessing from chronology alone.

4. Read Notebook 01 for conditional continuum reductions, with its historical sections interpreted through the final dossier and catalogue.

5. Before importing a result from the original raw account, check the baseline's exact version and source locations. Several proof-status statements require reconciliation.

Provenance and archive use

The examined repository is ethanwsussman/Yang–Mills-LLM-experiment, at commit

`e392ef45013a581664bab76b90867170b30fe10e`.

Its README explicitly reports an apparent $\mathbb{R}^4$ existence argument. The baseline report does not erase that claim; it examines the precise theorem scope and unreconciled status statements in the deposited files.

Our contributions are new to this project's record, not asserted novel relative to all literature. The bundle keeps proof, conditional input, open interface and superseded argument separate.

Each notebook directory contains a readable PDF, its compiling `main.tex`, and the exact assembled `original.tex`. The distribution source has only documented layout adjustments for paragraph wrapping and two long displays. Mathematical content is unchanged. Historical research files may contain old absolute workspace paths or old task-status comments; these are provenance, not current instructions.

Third-party primary articles and the source repository are linked and identified rather than republished. The provenance folder records the pinned tree and retrieval instructions. The reproduction folder documents building the delivered sources. The PDF compilation and visual inspection record states exactly what was checked.

The handoff is ready for mathematical review and continuation. Its most important open boundary is the passage from the controlled local construction to the complete multiscale Wilson density.